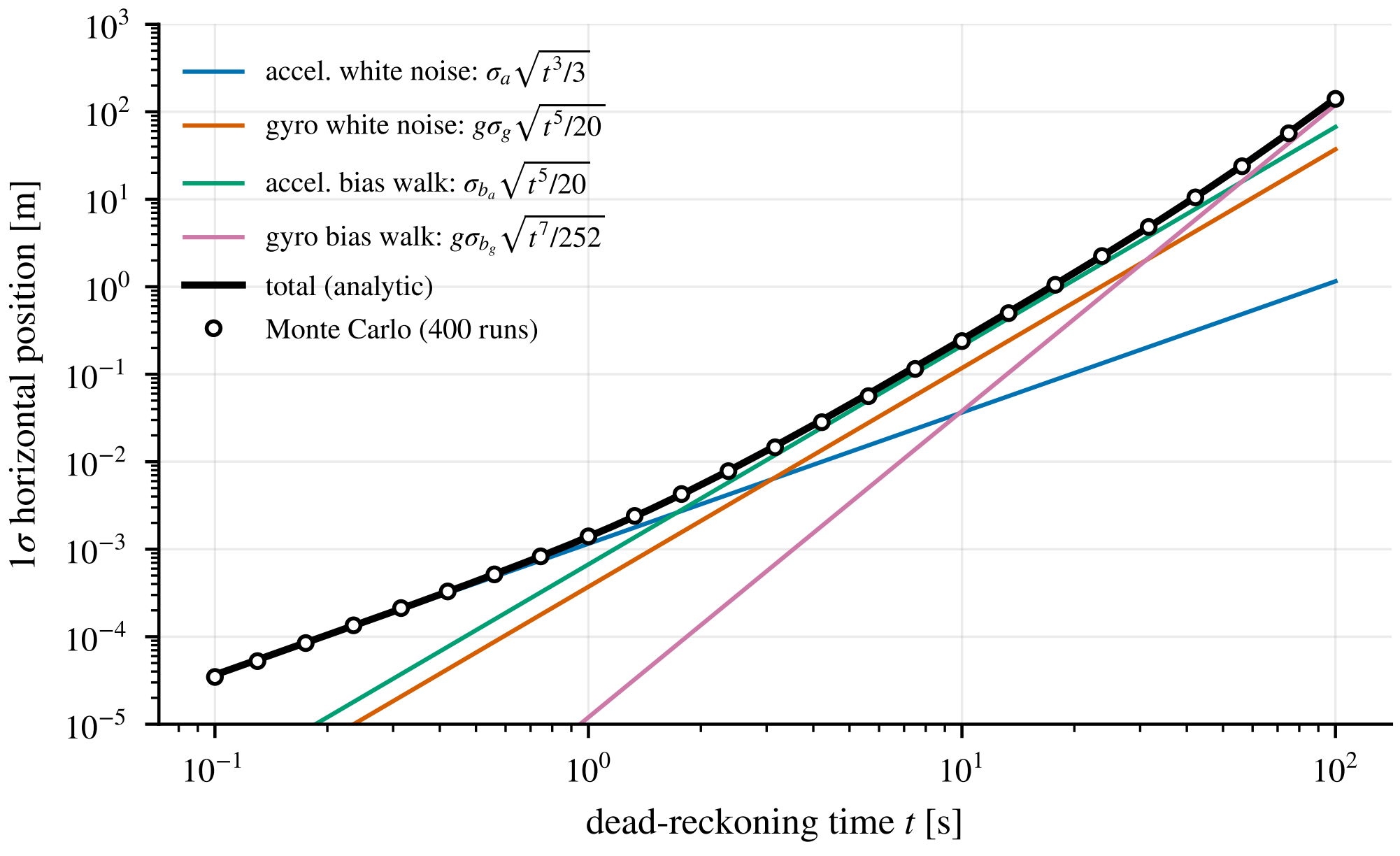
accel. white noise: $\sigma_a\sqrt{t^3/3}$
gyro white noise: $g\sigma_g\sqrt{t^5/20}$
accel. bias walk: $\sigma_{b_a}\sqrt{t^5/20}$
gyro bias walk: $g\sigma_{b_g}\sqrt{t^7/252}$
total (analytic)
Monte Carlo (400 runs)
1σ horizontal position [m]
dead-reckoning time t [s]
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-1}
10^{0}
10^{1}
10^{2}